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Artificial intelligence in urban forestry—A systematic review

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ABSTRACT

Environmental quality and the citizens' well-being largely depend on the urban forests. But managing this natural capital is challenging for its biological complexity and interactions with other environmental, social, and economic aspects of the cities. In line with the current digital revolution with the rise of Smart Cities, the use of Artificial Intelligence (AI) is becoming more common, including in urban forestry. In this systematic review, we evaluated 67 studies on the interplay between AI and urban forestry surveyed on Science Direct and Scopus to provide an overview of the state of the art and identify new research avenues. The sample includes studies in 23 countries and 85 cities, including 5 megacities, comprising the remote assessment of canopy cover and species distribution; ecosystem services assessment; management practices; and socioeconomic aspects of urban forestry. Most studies focused on extant urban forests, with few examples evaluating temporal trends, and only one focused on future scenarios despite the predictive potential of AI. A total of 22 AI methods were employed in these studies. Only half of them point to clear advantages of the chosen methods, such as robustness against missing data, overfitting, collinearity, non-linearity, non-normality, the combination of discrete and continuous variables, and higher accuracy. The choice of these methods depends on the various combinations of aim, timescale, data type, and data source. The application of AI in urban forestry is in full growth and will support decision making to improve livability in the cities.

1. Introduction

The promotion of the environmental quality in the cities is in accordance with the 2030 Agenda for Sustainable Development of the United Nations, more specifically the Sustainable Development Goal 11: Sustainable Cities and Communities (UN, 2015). This goal depends to a great extent on the provision of ecosystem services by the urban forests (De Jong et al., 2015). They include improving thermal comfort (e.g. Ferreira and Duarte, 2019), reducing air pollution (e.g. Escobedo and Nowak, 2009), increasing carbon sink (e.g. Nowak and Crane, 2002), supporting biodiversity (e.g. Livesley et al., 2016), to name a few. Because of these benefits, forest patches, parks, gardens, and street trees, which compose the urban forests in their broader sense (Konijnendijk, 2003), are the bases of Urban Greening, Green Infrastructure, and Nature-based Solutions (van den Bosch e Sang, 2017). But as for any natural system, forests are essentially complex in structure and function and this complexity tends to increase in the artificial urban landscape by its cultural, social, and economic aspects (Steenberg et al., 2015). Thus, planning and managing urban forests is a challenging and multidisciplinary task (Endreny et al., 2017).

Planning and management are core activities of urban forestry, a term that emerged in the United States in the late 19th century, but that only flourished in the 1960s (Konijnendijk et al., 2006). In the early 1990s, it also got broadly accepted in Europe (Konijnendijk, 2003) and later in the developing world (Schackleton et al., 2012). Urban forestry depends on a combination of theoretical and empirical knowledge and practices that first focused on tree management, but slowly came to consider the social and economic benefits of the urban forests (Konijnendijk et al., 2006). Urban forests governance depends on many activities: tree planting, maintenance, and survival assessment, surveys on habitat suitability, detailed inventories on canopy cover, species distribution (including invasive species); evaluation of their physical

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conditions (Pauleit and Duhme, 2000; Kenney et al., 2011; Hilbert et al., 2019), risk assessment and management (Martinez-Trinidad et al., 2010; Klein et al., 2019); and assessment of ecosystem services (Dobbs et al., 2011). Usually, these activities are based on laborious and time-consuming field campaigns undertaken by qualified staff (Alonzo et al., 2016). Such costly activities may only cover small areas in a given period and the evaluation of extensive areas overtime may be virtually impossible especially for the largest cities (McGee III et al., 2012; Alonzo et al., 2016). Eventually, remote inventories using satellite imagery (Li et al., 2020) and/or Light Detection and Ranging data (LiDAR, Alonzo et al., 2016) may be used, which are relatively cost- and time-efficient approaches, but still limited by accuracy issues.

Tackling this complexity of urban forestry is currently far from optimal, mostly because tools capable of integrating it were lacking in the past. As a response to the recent demand for more integrative tools, we saw the rise of a few platforms (Galle et al., 2019) including specialized modeling approaches such as the i-Trees to support decision making and management of urban forests (e.g. Kirnbauer et al., 2013; Hwang et al., 2017), but not free from limitations in species and spatial extrapolations (Parsa et al., 2019). Cutting-edge technologies that could deal with current urban forestry limitations have few references in state-of-art reviews to date (e.g. Bentsen et al., 2010; Roy et al., 2012; Maruthaveeran and van den Bosch, 2014; Ostoić and van den Bosch, 2015; Kabisch et al., 2015; Boulton et al., 2018; Song et al., 2018; Li et al., 2019; Lin et al., 2019; Ordóñez et al., 2019; Barona et al., 2020), and almost no reference to the applications of Artificial Intelligence (AI). This literature gap points out the niche avenues of the ever-growing use of artificial intelligence in various fields of knowledge (Kaplan and Haelein, 2019), urban forestry included.

Artificial intelligence consists of "a system ability to interpret external data correctly, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptations" (Kaplan and Haelein, 2019). This term was coined in the 1950s (Haenlein and Kaplan, 2019), and through moments of rise and fall, AI reached a point in the early 1990s in which computational power and data storage severely limited further development in the field (Muthukrishinan et al., 2020). A new rise of AI took place in the late 1990s when applications of deep-learning algorithms reached the public, and then in the 2010s when computer hardware limitations were overcome promoting further development of AI (Muthukrishinan et al., 2020). These algorithms may be divided into three families (whose boundaries are often unclear): discovering, learning, and reasoning (Contreras and Vehi, 2018). Discovering assumes that human interference may not be ideal to identify underlying patterns or information in large databases, including categorization of data displayed in complex scenarios and clustering techniques (Contreras and Vehi, 2018). Learning involves selecting techniques that allow computers to learn from simple filtering procedures to more complex deep-learning algorithms (Contreras and Vehi, 2018). Finally, reasoning stems from induction and deduction to generate heuristics and mimic human perception and decision (Contreras and Vehi, 2018). This power and flexibility to work with large datasets are fostering a revolution in data analyses in various applications.

The rapid expansion of artificial intelligence in the context of Smart Cities aims at improving the quality of life, but the natural capital of the cities has yet to be included in this revolution (Galle et al., 2019). The recent rise of the Internet of Nature and Smart Green Cities attests to the early application of class-leading technology in urban forestry (Clark and Cook, 2016; Galle et al., 2019) supported by the increasing trend in data availability from open-access digital platforms, monitoring programs combining sensors and Internet of Things, and volunteered information based on the citizens' science approach (Gulsrud et al., 2018; Nitoslawski et al., 2019). Such large and diverse datasets on urban forestry will likely benefit from powerful tools of AI (Chen et al., 2020) to integrate it with other environmental, social, and economic aspects of city governance (Galle et al., 2019). Whereas this combination of urban forestry and artificial intelligence has amassed a growing literature (Gulsrud et al., 2018), as AI slowly becomes democratized in research and development throughout the world (Montes and Goertzel, 2019), to date there is no comprehensive review of the interplay between urban forestry and artificial intelligence.

Artificial intelligence in urban forestry presents new challenges and opportunities for academia, practitioners, and policymakers. Since the application of AI in urban studies is in full growth, analyzing the intersections between AI and urban forestry and establishing a consistent mapping of the themes, techniques, and current limitations may provide the ground for further development. Thus, in this paper we endeavored to analyze the entire academic literature that covers both urban forestry and artificial intelligence to I) analyze the coherence in the historical and geographical evolution of urban forestry and its support by AI; II) assess the main urban forestry activities that have been supported by AI in science; III) evaluate the real benefits of AI compared to other standard statistical methods; VI) summarize the main methods of AI employed in this field; V) discuss the main advantages and limitations of the methods and their applications in urban forestry, VI) discuss the future steps for the development of AI in the field of urban forestry.

2. Material and methods

2.1. Study design

We used the PRISMA 2020 framework (Page et al., 2021) to survey and analyze the studies that combine artificial intelligence and urban forestry. Capturing the massive library of AI algorithms currently available can be challenging, not to mention the adaptations of standard methods and the development of new algorithms. This difficulty to access the related literature was clear from our first survey using the search expression ("urban tree" OR "urban forestry" AND "artificial intelligence"). This expression proved to be too inaccurate as these studies often referred to theoretical perspectives of artificial intelligence with no actual implementation of AI algorithms in data analyses. We also observed that studies using AI algorithms often did not refer to the term artificial intelligence, but rather specified the AI algorithms employed. Therefore, compiling a list of AI methods was mandatory for this systematic review.

An exhaustive list of AI methods is currently not available, and we had to compile it from different sources. Specific lists are available in studies from urban land dynamics (Wu and Silva, 2010), oil price forecasting (Sehgal and Pandey, 2015), oil industry (Rahmanifard and Plaksina, 2019), energy consumption (Daut et al., 2017), structural engineering (Salehi and Burgueno, 2018), image processing (Sadoughi et al., 2018), neurosurgery (Senders et al., 2018), groundwater modeling (Rajaee et al., 2019) and otolaryngology (Bur et al., 2019), bioinformatics (Ezziane, 2006), fracture mechanics (Nasiri et al., 2017), reservoir-hydro-environment (Allawi et al., 2018), agriculture (Jha et al., 2019), education (Zawacki-Richter et al., 2019), and pollution controls (Ye et al., 2020). In addition, summarized lists are also available from non-academic sources such as white papers, textbooks, and cataloging like Wikipedia.

Such lists may not capture the flexibility and potential tailoring of AI techniques to specific purposes. The emergence of widespread data-oriented programming languages (Python, Julia, and R) allowed scientists to keep improving methods and algorithms, making it virtually impossible to track all variants of the same method. As an example, there are so many modifications and additions to the initial artificial neural network methods that, unless one is using a default script or library, few studies will match the procedures. In addition, AI is currently going through unprecedented growth in use, acceptance, and investment (Muthukrishinan et al., 2020) that new methods or variations emerge every year. Such a list must also include other statistical algorithms that are not necessarily AI, but are still used in AI algorithms development. We used a list of AI methods and potential methods used in AI development in a search expression (refer to the supplementary material) on Web of Science and then validated on Scopus by comparing the results. Both platforms cover virtually all scientific production (Martín-Martín et al., 2018). From the final list of 643 articles, we excluded from the list all non-English articles, proceedings, and editorials (Fig. S1). We then assessed the records by four reviewers independently (the authors), and elected the potential studies. First, by reading the title and abstract and removing all studies not concerning directly urban forestry, and that clearly stated using non-AI methods. Then, by reading all the text to find the studies that employed AI in urban forestry.

2.2. Data collection and analyses

One reviewer compiled the data from the surveyed studies, which was then checked and discussed by all four reviewers. We obtained the data on geographical distribution, publication year, journal title, aims in urban forestry activities, the timescale of the studies, data source and type, AI methods, accuracy values (determination index - R^2, hit ratio, and relative root mean square error - rRMSE), and arguments supporting the choice of the AI methods. Data acquisition depended on their availability in each study.

We adopted a flexible approach by pre-defining some categories, while other categories emerged when reading the full texts. We used QGIS for analyzing geographical patterns, and R for data visualization using mainly radar graphics, histograms, and bar plots using the 'ggplot2' package (Wickham, 2016). We did not quantify the favorable arguments for each method as they could, or not, be presented specifically for each case study. Finally, we produced an alluvial plot using the 'easyalluvial' package (Koneswarakantha, 2021), which shows the main strategies adopted by the analyzed studies in terms of their aims, timescales, data type, data source, and main AI methods.

3. Results

Our systematic survey retrieved a final set of 67 research articles (full list of references in the Supplementary Information) from 23 countries (Fig. 1), out of which the United States (33 %) and China (22 %) are responsible for more than half of the studies. Other countries also stand out in the number of publications such as Canada (10 %), Finland (4%), and Germany (4 %). On a smaller geographical scale, a total of 85 cities have been studied in terms of urban forestry. Most of the studies, 80 %, focused on urban forestry in a single city, while 20 % of the studies evaluated urban forestry across multiple cities, from metropolitan areas to continental-wide scale. The studies included five megacities, with more than 10 million people, namely São Paulo in Brazil, and Beijing, Guangzhou, Shenzhen, and Hangzhou in China. Eight large cities with more than 5 million have also been studied, including Seoul in South Korea, London in the United Kingdom, New York City in the United States, Bogotá in Colombia, and Xuzhou, Nanjing, and Hong Kong in China. The first papers published in this sample are from the United States (Vidra et al., 2007) and China (Huang et al., 2007). The number of studies per year remained relatively low up to the early 2010s when the number of studies per year sharply increased (Fig. 2).

Studies were published in 36 outlets (Fig. S2). Almost a third of them consists of journals specialized in the urban environment such as 'Urban Forestry & Urban Greening' (17 %), 'Landscape and Urban Planning' (6 %), 'Urban Ecosystems' (5 %) and 'Urban Climate' (2%). Another share consists of journals specialized in remote sensing such as 'Remote Sensing' (9 %), 'Remote Sensing of Environment' (6 %), and 'International Journal of Remote Sensing' (5 %). The other half of the studies were published in journals with various scopes, including forestry: 'Forests', 'Forest Ecology and Management', 'Journal of Forestry Research', 'New Zealand Journal of Forestry Science'; or environment in its broad sense: 'Journal of Environmental Management', 'Environmental Geology', 'Modelling Earth Systems and Environment'; or outlets

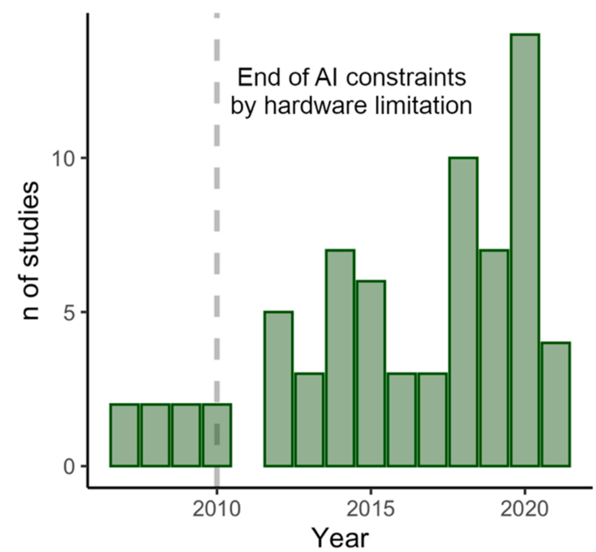

Fig. 2. Temporal variation in the number of studies on artificial intelligence and urban forestry. The dashed line indicates when AI hardware limitations were overcome, fostering the popularization of AI use in different fields.

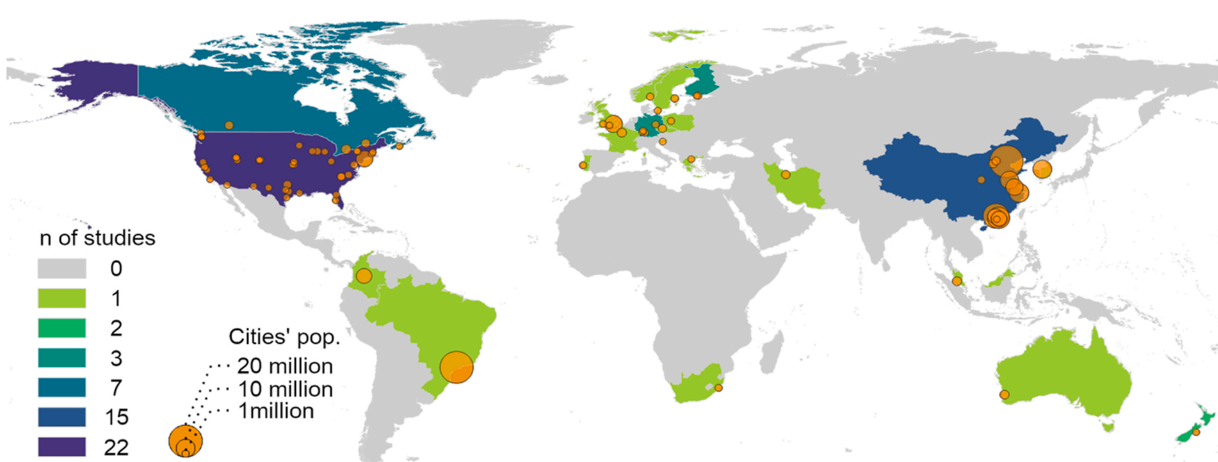

Fig. 1. Geographical distribution of the studies using artificial intelligence in urban forestry. The color gradient represents the number of studies by country and the circle size is proportional to the population of each city or metropolitan area.

specialized in computing science: 'Data', 'Environmental Modelling and Software' and the 'Journal of Supercomputing'.

We identified eight major categories of these applications that summarize the aims of the 67 studies (Fig. 3, Table S1). Most studies focused on the remote assessment of canopy cover (n = 15), remote classification of tree species (n = 14), remote forest, or land-use, classification (n = 7), and remote individual tree identification (n = 2). Other studies aimed at management activities (n = 10, including growth estimates, survival / mortality assessment, tree removal, vulnerability to insect outbreak), the evaluation of socioeconomic aspects of urban forestry (n = 9), and assessment of ecosystem services (n = 4). These aims were evaluated mostly in terms of the present conditions of the urban forests (n = 57), while nine studies evaluated the historical trends, and only one study endeavored to strive for future predictions from the datasets using AI methods.

Almost half of the studies relied on secondary data (n = 33) to achieve these aims (Fig. 4, Table S1), while 23 studies used solely primary data, and 11 studies used a combination of primary and secondary data. Remote sensors were the most used data source (n = 44). Studies also relied on field campaigns (n = 26), open-access datasets (n = 13), questionnaires or interviews (n = 5). Only one study obtained data from volunteered information platforms and none from local monitoring sensors. Overall, data were obtained at a city-wide scale including all units of the urban forests, namely: street trees, gardens, parks, and forest patches (n = 20). Some studies focused solely on forest patches (n = 16), parks (n = 1), gardens (n = 7), or street trees (n = 9), or a combination of the mentioned units (n = 21). In these units of the urban forests, more

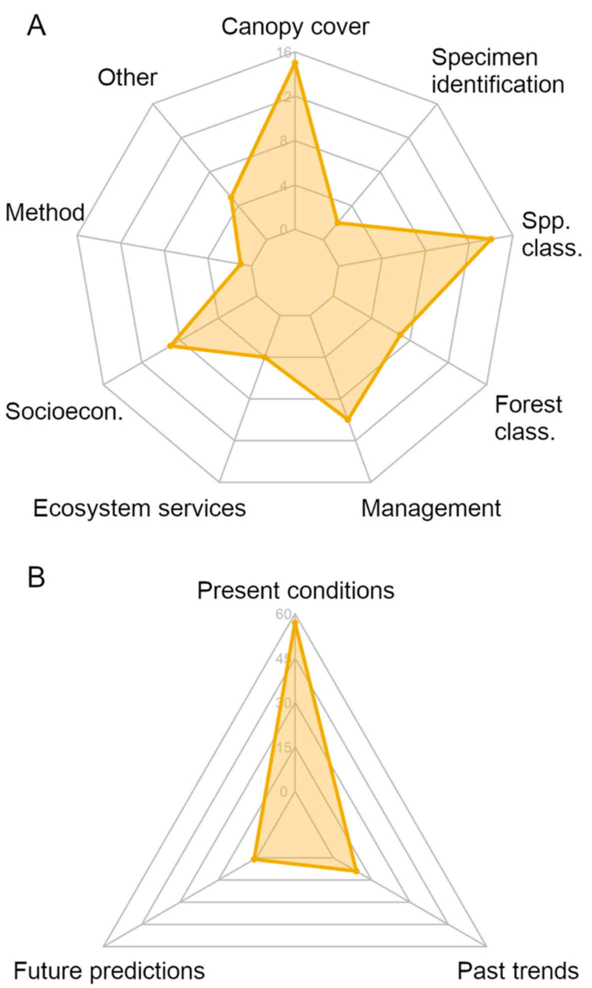

Fig. 3. Overview of A) the aims of the studies that employed AI in urban forestry, and B) their timescales.

than half of them studied variables related to forests canopy or the crown of the trees (n = 39), followed by species composition (n = 11) and diameter at breast height (DBH, n = 10). Any of the analyzed studies evaluated the below-ground aspects of the urban forests.

Data analyses relied on a total of 22 AI methods. About 80 % of the studies used one or a combination of the following methods, Random Forests, Support-vector Machine, Classification and Regression Trees, Artificial Neural Networks, Hierarchical Clustering Analysis, Nonmetric Multidimensional Scaling, Convolutional Neural Networks, Discriminant Analysis, K-means Clustering, or k-nearest Neighbors (Fig. 5, Table S1). The remaining 12 algorithms were used by single studies for specific purposes. There is an overall predominance in learning, supervised, algorithms, and no example of purely reasoning AI algorithms. The overall accuracy of these AI methods in the predictions of different features in urban forests (Fig. 6) can be as high as 96.4 % (R^2 or hit ratio), or 10.1 % in terms of relative root mean square error (rRMSE). Three studies reported overall accuracies (R^2 or hit ratio) equal to or lower than 50 %. For the studies that compared AI and standard statistical methods, there is an overall increase in the accuracy by using AI methods, as high as 68.3 %. Still, one study shows a weaker performance of AI methods than other standard statistical methods, reporting a 31.8 % reduction in prediction accuracy.

Given the large number of AI methods used by the studies, we also evaluated the reported arguments for the choice of methods (Fig. 7). More than half of the studies (n = 36) provided the technical advantages of the chosen AI methods. A fourth of the total relied on the successful use of the methods by previous studies, and a relatively high number of studies in the sample (14) gave no details on the choice of the AI methods. Despite the reasons given by the authors, Fig. 8 shows that the choice of the most used AI methods depends on the combination of the aims of the study, timescale, data type, and data sources. For instance, remote species classification relied on Support-vector Machine, Random Forests, Classification and Regression Trees, and Artificial Neural Networks. Tree identification over a historical perspective relied mainly on Random Forests. The only study concerning future scenarios for canopy cover over the city used Random Forests for the predictions. On the other hand, most studies concerning urban forest management evaluated the current conditions of the forest based on field campaigns using mainly Non-metric Multidimensional Scaling or Artificial Neural Networks, or alternatively Hierarchical Clustering to analyze their datasets. Concerning the data source, questionnaire and interviews were almost exclusively used in socioeconomic studies, while open-access datasets were mainly used by forest classification studies, but also by management and canopy cover studies. Datasets based on both field campaigns and remote sensors were those requiring ground truth. They include remote species classification, individual tree identification, and forest classification.

4. Discussion

The spatiotemporal distribution of the studies in the sample reflects the development of both urban forestry and AI. The United States, where the term urban forestry was coined, concentrates a third of all studies. This protagonism in the field of urban forestry also holds true for studies concerning urban tree benefits and costs (Song et al., 2018), urban forest governance (Ordóñez et al., 2019), and human interactions in urban green spaces (Kabisch et al., 2015). Europe comes next by adopting the term urban forestry only later, and by representing a fourth of the studies. Most of the other studies come from China and other developing countries. In terms of timeline, AI in urban forestry showed a late development compared to the trends revealed by other systematic reviews on urban forestry (Kabisch et al., 2015; Boulton et al., 2018; Ostoić and van den Bosch, 2015; Lin et al., 2019; Ordóñez et al., 2019; Barona et al., 2020). Such late development coincides with AI reaching a wider public and the end of hardware limitations by 2010 (Muthukrishinan et al., 2020).

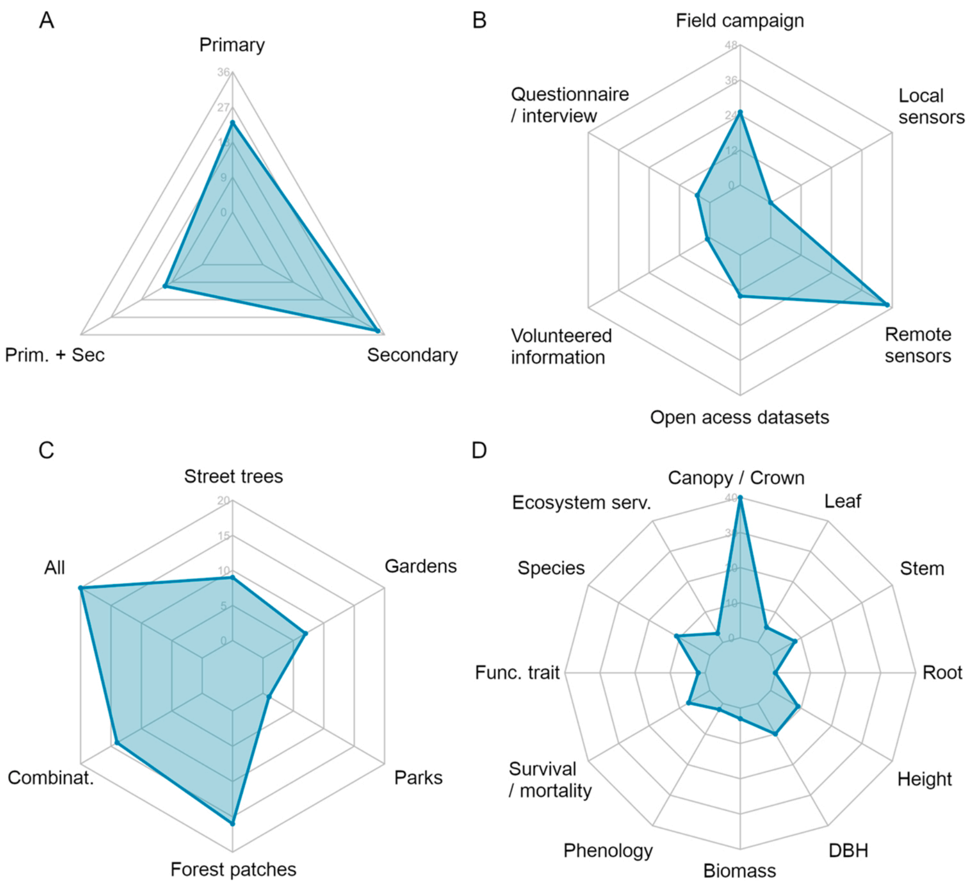

Fig. 4. Characteristics of the datasets used in the studies concerning AI and urban forestry, in terms of the number of studies. A) Data source, whether data used were produced in the context of the studies, primary data, or obtained from available datasets, secondary, or a combination of both. B) Methods of data acquisition, studies may combine more than one method. C) Sample unit in urban forestry, studies may combine more than one sample unit. D) Variables measured in the analyzed studies, which may combine more than one variable.

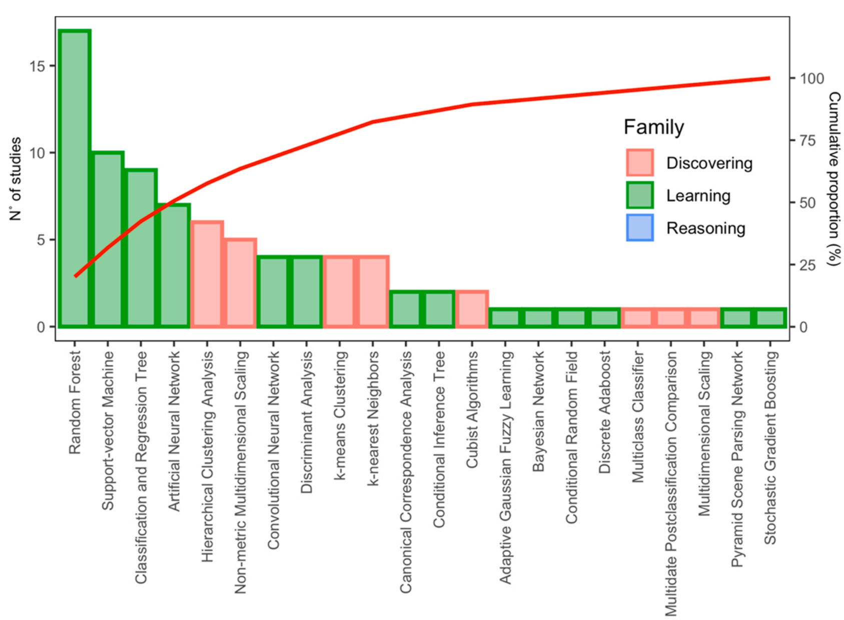

Fig. 5. Pareto chart of all AI methods used in the studies from the analyzed sample. Methods are discriminated by families according to their applications (modified from Contreras and Vehi, 2018). The red line indicates the cumulative proportion of studies. Surveyed studies may combine more than one method of AI.

4.1. Applications of AI in urban forestry research

Artificial Intelligence found its early applications in urban forestry in the field of remote sensing (e.g. Huang et al., 2007; Yuan, 2008; Walton, 2008) as planning and management of urban forests could no longer rely solely on field campaigns (Alonzo et al., 2014). This application was only possible because satellite images were increasingly available, at higher spatial and spectral resolution (Loveland, 2012). The classification of satellite images is a sensitive step (Mountrakis et al., 2011) and it is the main application of AI in urban forestry to date (e.g. Walton, 2008;

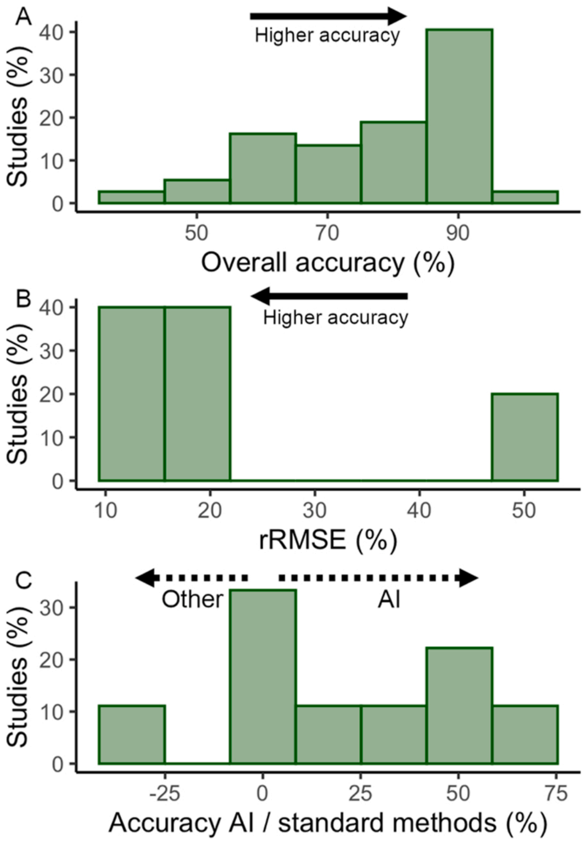

Fig. 6. Variation in the reported accuracies by the studies using AI in urban forestry. A) Histogram of the overall variation in accuracy values as reported as R^2 or hit ratio (n = 37). B) Histogram of the relative root mean square error (rRMSE), lower values indicate higher accuracy (n = 5). C) Ratio between the accuracies from AI methods and standard statistical methods used to analyze the same datasets (n = 9).

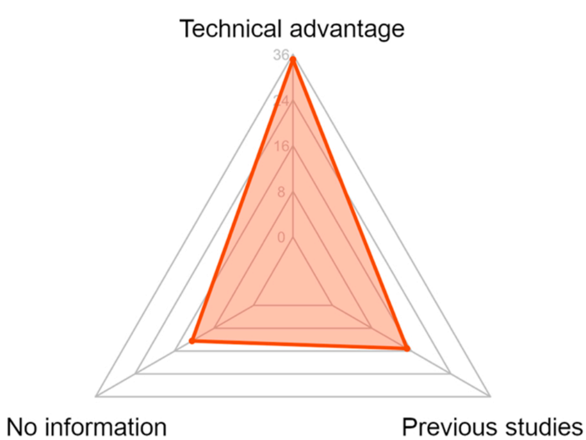

Fig. 7. Arguments given by the studies for choosing an AI method such as technical details on the advantages, or if they rely solely on previous studies' success, or if they do not provide any detailed information on the choice of the AI method.

McGee III et al., 2012; Mincey et al., 2013; Zhang and Hu, 2014; Wang et al., 2016; Franco and McDonald, 2018; Haase et al., 2019; Wagner and Hirye, 2019; Shen et al., 2020). In addition to the automatic recognition of vegetation cover, and remote identification of individual trees (e.g. Tanhuanpää et al., 2014), multi- and hyperspectral images allowed the remote recognition of tree species, which is fundamental information for urban forestry (Ardila et al., 2012; Huang et al., 2013; Pu et al., 2018; Roffey and Wang, 2020), including the control of invasive species (Singh et al., 2015). Both the assessment of vegetation cover and species identification further benefited from the massive data produced by the laser imaging detection and ranging systems (LiDAR), often from open-access products (Zhang and Qiu, 2012; Verlič et al., 2014; Zhang et al., 2016; Shojanoori et al., 2018; Liu and Wu, 2018; Baines et al., 2020; He et al., 2020). The classification procedure largely depends on the ability of AI to strive for patterns from large datasets. The AI methods yielded consistently higher accuracy values compared to other standard statistical methods in the studies. However, this ability of AI in pattern recognition tends to decrease when predicting an increasing number of forest classes or species (e.g. Fang et al., 2020). More recently, AI supported the use of multispectral or RGB high-resolution images from unmanned aerial vehicles to evaluate the distribution of trees and their species in the city (Lin et al., 2015; Minarík et al., 2020; Zhang et al., 2020; Wang et al., 2021). This application consists of a cost-efficient alternative to produce such fundamental data for proper planning and management of the urban forests (McGee III et al., 2012).

Further development in imaging systems allowed the evaluation of vegetation from a ground-level perspective. Street-level assessment of the urban forests is now possible with more than a decade of image capturing and processing by Google Street View (Anguelov et al., 2010). The vegetation perceived by the pedestrians is evaluated using the Green View Index, in other words, the proportion of pixels representing a tree cover in the image (Li, 2020). Identifying the tree canopy in RGB images can be challenging when compared to multi- and hyperspectral images, and AI has a central role in extracting the canopy pixels. Ground-level imaging has been used for estimating the spatial distribution of perceived canopy cover in many cities around the world (e.g. Seiferling et al., 2017; Stubbings et al., 2019), which is a fundamental factor in promoting citizens' perceived well-being (Kothencz et al., 2017).

While the remote assessment of urban forests and species distribution was largely supported by AI, a more in-depth evaluation of the structure and dynamics of the urban forest has consistently lagged behind. The scope diversity of the 36 outlets in this sample reflect the great potential of AI in various activities of urban forestry. These activities include the support to estimate missing data common to field surveys on tree size (Diamantopoulou, 2010) or to estimate tree' dimension, including height and DBH, over the entire city using LiDAR data (Saarinen et al., 2014; Baines et al., 2020; Li, 2020). Tree dimension is a key feature related to ecosystem services (Sun et al., 2015; Xu et al., 2020), such as temperature reduction (Gage and Cooper, 2017) and carbon storage (Strohbach and Haase, 2012; Zhang and Shao, 2021). The evaluation of other ecosystem services like the reduction of air pollution has also been supported by the implementation of AI (Lee et al., 2021). Planting, growth, survivorship of trees, and assessment of species distribution (Jutras et al., 2010; BenDor et al., 2014; Roman et al., 2014; Zhang and Jim, 2014; Jim and Zhang, 2015; Morgenroth et al., 2017; Nitoslawski et al., 2017; Elmes et al., 2018; Guo et al., 2018; Koch et al., 2018; Steenberg, 2018), including invasive ones (Mavimbela et al., 2018; Dyderski and Jagodziński, 2019), are other key aspects of urban forestry supported by AI using primary and / or secondary datasets on biophysical, maintenance and socioeconomic features. This combination may again lead to higher accuracy in AI predictions, incuding the predictions of real estate valuation in relation to the green infrastructure (Yoo et al., 2012) when compared to other statistical methods. Questionnaires and interviews are other sources of valuable primary information to evaluate the perceived importance of the urban forest by different stakeholders using AI (Escobedo et al., 2020; Baumeister et al., 2020; Franco and McDonald, 2018).

Despite the successful applications of AI in urban forestry demonstrated by the analyzed studies, this systematic review detected gaps in studies timescale and data source. Most studies focused on describing extant conditions of urban forests, canopy cover, and species distribution. Few studies evaluated past trends in urban forestry, mostly based

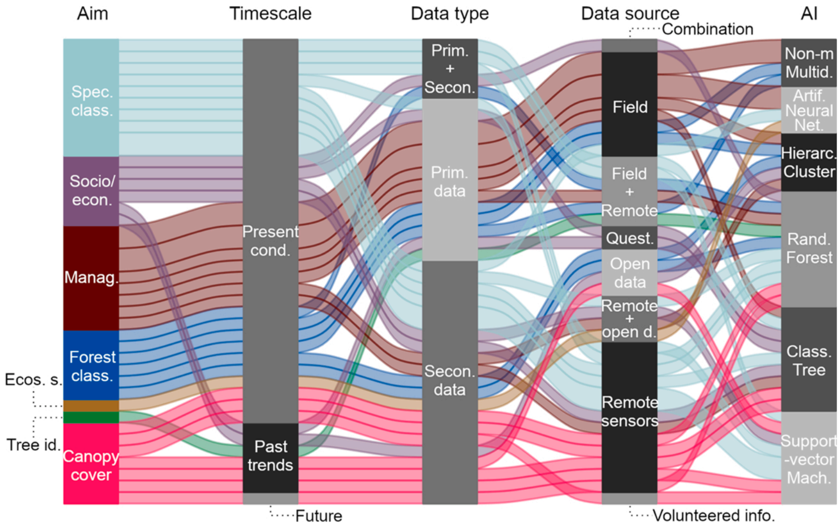

Fig. 8. Summary of the main aims of the sutdies represented by different colors (Socio/econ. – socioeconomics, Spec. class – species remote classification, Manag. – management practices, Tree id. – remote tree identification, Forest class. – remote classification of urban forests, Ecos. S. – ecosystem services, Canopy cover – remote assessment of canopy cover); in addition to timescales (Present cond. – present conditions, Histor. trends – historical trends, Future – future predictions); data type (Prim. data – primary data, Secon. data – secondary data); data source (Remote sensors – including satellite imagery, LiDAR, and other, Open data – Open acess datasets, Field – field campaigns, Quest. – questionnaire and interviews) and main AI methods (Support-vector Mach. – Support-vector Machine, Rand. Forests – Random Forests, Non-m. Multid. – Non-metric Multidimensional Scaling, Hierac. Cluster – Hierarchical Clustering, Class. Tree – Classification and Regression Trees, Artif. Neural Net. – Artificial Neural Networks).

on satellite or ground-level imagery (e.g. Sun et al., 2009; Lin and Wang, 2021). Despite the potential datasets available from the Smart Cities, only one study took advantage of this digital revolution of the cities to obtain data from volunteered information, more specifically OpenStreetMap (Griffith and Hay, 2018), and none from local monitoring to evaluate the past and present conditions of urban forests. Future predictions also remained poorly explored despite the potential of AI algorithms to strive for scenarios from various datasets to support decision making on urban forest governance (Saunders et al., 2020). Finally, there are limited applications of AI in estimating forests' biomass (e.g. Sun et al., 2015; Zhang and Shao, 2021), assessing plants' phenology (e.g. Huang et al., 2013) and functional traits (e.g. Gu et al., 2015; Mosaffaei and Jahani, 2020), and no applications concerning the below-ground organs of trees.

4.2. AI methods

All these applications in urban forestry reseach relied on a total of 22 AI methods. This high number of methods partially demonstrates the success of using the proposed search expression in this systematic review. Nonetheless, it may have not captured tailored-made methods given their diversity and fast development (Muthukrishinan et al., 2020). About 80 % of the studies in the sample used only eight AI methods. These algorithms require less programming and technical know-how, except for artificial networks and their variants (e.g. Jutras et al., 2010; Lin et al., 2020). All 22 algorithms belong to either discovering or learning families and focus on the association among different variables, classification, and clustering (Contreras and Vehi, 2018). No study employed purely reasoning methods, probably because of the nature of the analyzed variables. Nonetheless, the few studies based on questionnaires and interviews (e.g. Ramage et al., 2013; BenDor et al., 2014; Gerstenberg and Hofmann, 2016; Baumeister et al., 2020; Escobedo et al., 2020) could benefit from fuzzy logic and natural language processing techniques for text interpretation. This methodological gap points to new avenues in the use of more complex applications of AI in the interpretation of questionnaires, and other applications like compared legislation, automatic analysis of scientific outlet databases, internet activism, to name a few.

It is also worth noticing that AI does not always stand alone in the studies, rather a significant part of the studies uses AI as a part of their main workflow. These can be divided into two subgroups according to their strategy: I) studies based on the combination of AI techniques, and II) studies that combine AI with a traditional statistical method. In the first group of studies (e.g. Pu, 2009; Yoo et al., 2012; Pu and Landry, 2012; BenDor et al., 2014; Zhou et al., 2019; Zhang et al., 2020), the main concerns were testing whether a set, or sequence, of techniques is appropriate to perform a specific analysis or when each technique in the workflow provides a specific result. For example, Classification and Regression Trees produce highly interpretable results useful for decision making. But its intrinsic high model variance (De'Ath and Fabricius, 2000) calls for complementary methods such as Random Forests (Saunders et al., 2020), or Bootstrap Aggregating methods (i.e. Bagging, Escobedo et al., 2020), or others, to produce reliable results. Alternatively, combining algorithms allows testing which one of a set of techniques might yield the best results. This comparison follows a traditional path in computational science to test the best algorithms for a specific scenario. In the second group, AI combined with other statistical methods usually occurs when complementarity is sought. For instance, the results from an AI application may be fed into other statistical analyses (e.g. Elmes et al., 2018; Franco and McDonald, 2018). Or, they can support traditional modeling platforms in the field such as i-Tree Eco (Cimburova and Barton, 2020) for more precise spatial extrapolations of its outputs.

A key question then is why given AI methods were chosen in each study. About half of the urban forestry studies provide clear technical arguments in favor of the chosen AI methods. These arguments include the potential for higher prediction accuracy; capacity to predict multiple response variables; ability to deal with data from various sources common to urban forestry studies; ability to combine discrete and continuous variables; robustness against missing data, overfitting, non-linearity, non-normality, and collinearity. Still, 20 % of the studies gave no arguments in favor of the chosen AI methods. Thus, AI use must be carefully planned in the studies on urban forestry, as these techniques may not be appropriate or necessary for all studies. Examples such as Zhou et al. (2019) and Wang et al. (2019) showed that the most complex algorithms do not necessarily yield the best results. This broad range of possibilities of urban forestry research aims, timescale, data source and type, and AI methods is evident in Fig. 8. As such, researchers need to provide the appropriate methodological foundations for the chosen algorithms, so that the use of AI is not seen as excessive and unjustifiable.

4.3. AI in future decision making

The capacity of AI algorithms to analyze large and complex datasets, and yield comprehensive results, is already supporting effective and cost-efficient decision making in both private and public sectors (i.e. Duan et al., 2019; Valle-Cruz et al., 2020), where, more than automating decision making, AI is expected to support and augment decision making by humans (Duan et al., 2019). The present systematic review shows that the applications of AI in urban forestry, although in full growth, are mainly focused on specific research questions mostly related to the characterization of the urban green spaces, which is a significant step in terms of urban forestry management. Nonetheless, there is still room for improving AI applications, methods, and workflows to better support decision making in urban forestry. Based on the present review we list five propositions for future development in the field.

Proposition 1. Dealing with various types of datasets is one of the main strengths of AI methods (De'Ath and Fabricius, 2000). Thus, decision making would greatly benefit from multidisciplinary studies that combine soft and hard science to strive for more holistic patterns from the urban forests and their interaction with society.

Proposition 2. Urban forestry may strongly benefit from the use of reasoning AI methods. Fuzzy logic and natural language processing, among others, could support the evaluation of the effectiveness of decision making through the analyses of legal instruments, policies, scientific papers, social media, and others.

Proposition 3. The ability of AI for striving pathways and multiple future scenarios (e.g. Saunders et al., 2020) brings a unique opportunity for leveraging the engagement and discussion among different actors of the society to promote participative decision making in urban forestry.

Proposition 4. Most AI methods and tools still require specialized knowledge and skills (Duan et al., 2019) not always found in the public departments precluding the independent use and application of these methods. Research should focus on more friendly platforms that are constantly fed and trained by field data for more dynamic and adaptive decision making. New technologies, tools and/or paradigms, such as OpenML (open machine learning) and the adoption of low-code / point-and-click tools, may help public bodies to adopt more often and more efficiently AI in their decision-making processes.

Proposition 5. Many AI algorithms focus on yielding the highest predictive power at the cost of interpretability, but it is still important to improve transparency and further develop explainable AI (XAI) algorithms that explicitly reveal underlying causes (Miller, 2019) so that decision makers learn from the analyses and develop better policies.

Such propositions will only come to reality if there is a significant change in the culture of technology acceptance by decision makers (Duan et al., 2019), and if sufficient resources are allocated for further development of AI in urban forestry that not rarely lacks financing and political prioritization (Escobedo et al., 2011).

5. Conclusions

This systematic review brought to light the great potential of using AI methods in the various activities that characterize urban forestry. A total of five megacities, and other 8 cities with more than 5 million people, were analyzed proving the potential of AI in urban forest planning and management across extensive and complex areas. Despite these successful applications, this review also detected key knowledge gaps that may open future research avenues. Regarding the urban forestry practices or aims of the studies, there is still significant room for studying intrinsic characteristics of the tree species such as phenology, functional traits, and belowground organs of trees, in addition to species ecosystem services and related biological mechanisms, and socioeconomic aspects of urban forestry. There is also the almost unexplored potential of using AI for predicting future scenarios in urban forestry, essential for supporting decision making and governance. Regarding the datasets for analyses, rich data from local sensors produced in the context of the Smart Cities and even volunteered information are becoming increasingly available to be explored by urban forestry research with the support of AI methods.

It is important to highlight that the choice of the AI methods may be challenging given the large number of algorithms currently available, and the rapid development of new algorithms. Thus, we advise researchers in the field of urban forestry to provide technical details and arguments to support their choices. The AI methods should not be considered as silver bullets to promote the studies. Despite this remark, there is a broad range of possibilities combining urban forestry practices, timescale, data, and AI methods to support narratives and decision making.

We finally present five propositions for leveraging AI in decision making in the field of urban forestry, I) multidisciplinary studies combining soft and hard science, II) application of reasoning methods for assessing the effectiveness of decision making, III) striving pathways and scenarios to support participative decision making, IV) improving methods using friendly platforms, V) and improving transparency through the development of AI algorithms that explicitly reveal underlying processes.

Author contribution statement

HCLA, GML: conceived the idea, HCLA, FSM, TT, GML designed the study. FSM: obtained the data. HCLA, FSM, TT, GML: analyzed the data. HCLA, FSM, TT, GML: wrote the paper.

Declaration of Competing Interest

The authors declare no conflict interest.

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Appendix A. Supplementary data

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